

Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2026

Course: B.Tech
Code: 100202

Subject: Engineering Mathematics II

Time: 03 Hours
Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):

[2 x 7 = 14]

- (a) The harmonic conjugate of the function $u(x, y) = x^2 - y^2$ is
 - (i) $2xy$
 - (ii) $-2xy$
 - (iii) $x^2 + y^2$
 - (iv) xy
- (b) If a function is analytic inside and on a simple closed contour C , then $\oint_C f(z) dz$
 - (i) $2\pi i$
 - (ii) 0
 - (iii) ∞
 - (iv) Depends on the contour
- (c) The residue of the function $f(z) = \frac{e^z}{z}$ at $z = 0$ is
 - (i) 0
 - (ii) e
 - (iii) 1
 - (iv) $2\pi i$
- (d) The auxiliary equation of $y'' - 5y' + 6y = 0$ is
 - (i) $m^2 + 5m + 6 = 0$
 - (ii) $m^2 - 5m + 6 = 0$
 - (iii) $m^2 - 6m + 5 = 0$
 - (iv) $m^2 + 6m - 5 = 0$
- (e) The standard form of a Cauchy-Euler differential equation is
 - (i) $y'' + py' + qy = r(x)$
 - (ii) $x^2 y'' + axy' + by = 0$
 - (iii) $y'' + x^2 y = 0$
 - (iv) $y' + xy = 0$
- (f) The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if
 - (i) $p < 1$
 - (ii) $p = 1$
 - (iii) $p > 1$
 - (iv) $p \leq 1$
- (g) Cauchy's Condensation Test can be applied to
 - (i) Any alternating series
 - (ii) Any finite series
 - (iii) Positive, decreasing series
 - (iv) Oscillating series
- (h) The Laplace Transform of 2^t is
 - (i) $\ln 2$
 - (ii) $\frac{1}{s-2}$
 - (iii) $\frac{1}{s - \ln 2}$
 - (iv) $\frac{1}{s}$
- (i) The inverse Laplace Transform of $\frac{1}{s^2+9}$ is
 - (i) $\sin 3t$
 - (ii) $\frac{1}{3} \sin 3t$
 - (iii) $3 \sin 3t$
 - (iv) $\cos 3t$
- (j) The half-range cosine series is used when the function is extended as a
 - (i) Periodic function
 - (ii) Odd function
 - (iii) Even function
 - (iv) Constant function

- Q.2** (a) Prove that the function $f(z) = z^3$ is analytic. [7]
 (b) Show that the function $u(x, y) = e^x(x \cos y - y \sin y)$ is harmonic. Hence, find its harmonic conjugate $v(x, y)$ and obtain the corresponding analytic function $f(z)$. [7]
- Q.3** (a) Evaluate the contour integral $\oint_C \frac{2z+1}{(z-1)(z-2)} dz$, where C is the circle $|z-1|=2$, using the Cauchy Integral Formula. [7]
 (b) Find the Laurent series expansion of the function $f(z) = \frac{1}{z(z-1)}$ valid in the region $0 < |z| < 1$. Hence, determine the nature of the singularity at $z = 0$. [7]
- Q.4** (a) Evaluate the definite integral $\int_0^{2\pi} \frac{d\theta}{5+4 \cos \theta}$, using the method of residues. [7]
 (b) Solve the differential equation $y'' - 4y' + 4y = xe^{2x}$. [7]
- Q.5** (a) Solve the Cauchy-Euler equation $x^2y'' + 5xy' + 4y = x^3$. [7]
 (b) Using the Method of Variation of Parameters, solve the differential equation $y'' + y = \sec x$. [7]
- Q.6** (a) Test the convergence of $\sum_{n=1}^{\infty} \frac{2^n}{n!}$ using D'Alembert's Ratio Test. [7]
 (b) Apply Raabe's Test to determine the nature of $\sum_{n=1}^{\infty} \frac{(2n)!}{4^n(n!)^2}$. [7]
- Q.7** (a) Find the inverse Laplace Transform of $(s) = \frac{2s+5}{(s-1)(s^2+4)}$. [7]
 (b) Solve the initial value problem using the Laplace Transform $y'' + 5y' + 6y = e^{-2t}$, where $y(0) = 1$, $y'(0) = 1$. [7]
- Q.8** (a) Find the Fourier series of the function $f(x) = x$, where $-\pi < x < \pi$. Hence, deduce the value of $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$. [7]
 (b) Obtain the half-range sine series of $f(x) = x(\pi - x)$, where $0 < x < \pi$. [7]
- Q.9** Write short note on **any two** of the following: [7 x 2 = 14]
 (a) Cauchy-Riemann Equation
 (b) Laurent's Series
 (c) Convolution Theorem
 (d) Parseval's theorem